Addition and subtraction of rational numbers together

Long chains of fractions — one common denominator, one pass.

LESSON 43 | 1 × 40 MIN | MATEMATIKA 7, §18 | S8 7.2

LESSON CLOCK

10' | 12' | 12' | 6'

One denominator for all	10 min
Mixing forms	12 min
Simple equations	12 min
Homework	6 min

LEARNING OBJECTIVES

- Evaluate a chain of additions and subtractions of rational numbers.
- Use one common denominator for the whole chain.
- Combine fractions and decimals in one calculation.
- Solve a simple equation of the form x + a = b in $\mathbb{Q}$.

TEXTBOOK

National	pp. 114–117
Cambridge	Stage 8, pp. 70–76
Workbook	Exercise 7.2

01 Explanation

One denominator for all

For a chain such as $\frac{1}{2} - \frac{2}{3} + \frac{3}{4}$, find the LCM of **all** the denominators at once and rewrite everything.

$$\frac{6}{12} - \frac{8}{12} + \frac{9}{12} = \frac{7}{12}$$

CHAIN	LCM	ANSWER
$\frac{1}{2} - \frac{2}{3} + \frac{3}{4}$	*12*	$\frac{7}{12}$
$\frac{2}{5} + \frac{1}{2} - \frac{7}{10}$	*10*	$\frac{1}{5}$
$-\frac{1}{3} - \frac{1}{4} + \frac{5}{6}$	*12*	$\frac{1}{4}$

ONE CONVERSION, NOT TWO

Working pairwise means converting twice and simplifying twice. Converting the whole chain once is shorter and far less error-prone.

Mixing forms

When fractions and decimals appear together, convert to whichever form is easier.

EXPRESSION	BEST FORM	ANSWER
$0.5 + \frac{1}{4}$	fractions	$\frac{3}{4}$
$0.25 - \frac{1}{3}$	fractions	$-\frac{1}{12}$
$\frac{1}{2} + 0.3$	decimals	0.8
$\frac{1}{3} + 0.2$	fractions	$\frac{8}{15}$

USE FRACTIONS WHENEVER A THIRD IS INVOLVED

$\frac{1}{3}$ has no exact decimal form, so any decimal working is an approximation. Fractions keep the answer exact.

Simple equations

$x + a = b$ is solved by subtracting a from both sides:

$$x = b - a$$

EQUATION	WORKING	ANSWER
$x + \frac{1}{3} = \frac{5}{6}$	$\frac{5}{6} - \frac{2}{6}$	$\frac{1}{2}$
$x - \frac{2}{5} = \frac{1}{10}$	$\frac{1}{10} + \frac{4}{10}$	$\frac{1}{2}$
$x + \frac{3}{4} = \frac{1}{4}$	$\frac{1}{4} - \frac{3}{4}$	$-\frac{1}{2}$

ALWAYS SUBSTITUTE THE ANSWER BACK

One line of checking catches every sign error and every denominator slip, and the check is quicker than the solution.

02 Worked examples

EXAMPLE 1 **Compute $\frac{1}{2} - \frac{2}{3} + \frac{3}{4}$.**

1. LCM of $2, 3, 4$ is 12.
2. $\frac{6}{12} - \frac{8}{12} + \frac{9}{12}$
3. $= \frac{6 - 8 + 9}{12}$
4. $= \frac{7}{12}$

ANSWER

$\frac{7}{12}$

EXAMPLE 2 **Compute $0.25 - \frac{1}{3}$.**

1. $0.25 = \frac{1}{4}$

   Fractions are exact.

2. LCM of 4 and 3 is 12.

3. $\frac{3}{12} - \frac{4}{12}$

4. $= -\frac{1}{12}$

ANSWER

$-\frac{1}{12}$

EXAMPLE 3 **Solve $x + \frac{3}{4} = \frac{1}{4}$.**

1. $x = \frac{1}{4} - \frac{3}{4}$

2. $= -\frac{2}{4}$

3. $= -\frac{1}{2}$

4. Check: $-\frac{1}{2} + \frac{3}{4} = \frac{1}{4}$ ✓

ANSWER

$x = -\frac{1}{2}$

03 Interactive model

Do one chain twice on the board — pairwise and all at once — and let the class time both; the single conversion wins visibly.

One denominator for the whole chain

DENOMINATOR	FACTORISED	EXTRA FACTOR NEEDED
2	2	6
3	3	4
4	2^2	3

LOWEST COMMON DENOMINATOR

$$12$$

$$\frac{6}{12} + \frac{8}{12} - \frac{9}{12} = \frac{5}{12}.$$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Chain of operations	Amallar zanjiri	Цепочка действий
Common denominator	Umumiy maxraj	Общий знаменатель
To combine	Birlashtirish	Объединить
Equation	Tenglama	Уравнение
Unknown	Noma’lum	Неизвестное
Check	Tekshirish	Проверка
Decimal form	O‘nli ko‘rinish	Десятичная форма
Simplest form	Eng sodda ko‘rinish	Несократимый вид

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

For a chain, use the LCM of:

- the first two
- all the denominators
- the largest
- the smallest

$\frac{1}{2} - \frac{2}{3} + \frac{3}{4}$ equals:

- $\frac{7}{12}$
- $\frac{5}{12}$
- $\frac{11}{12}$
- $-\frac{7}{12}$

$0.25 - \frac{1}{3}$ equals:

- $\frac{1}{12}$
- $-\frac{1}{12}$
- -0.08
- $\frac{7}{12}$

When a third is involved, prefer:

- decimals
- fractions
- either
- percentages

$x + \frac{3}{4} = \frac{1}{4}$ gives:

- 1
- $\frac{1}{2}$
- $-\frac{1}{2}$
- -1

After solving you should:

- stop
- substitute back
- round
- draw a graph

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $\frac{1}{2} + \frac{1}{3}$

ANSWER $\frac{5}{6}$

2. $\frac{1}{2} - \frac{1}{3}$

ANSWER $\frac{1}{6}$

3. $0.5 + \frac{1}{4}$

ANSWER $\frac{3}{4}$

4. $\frac{1}{2} + 0.3$

ANSWER 0.8

5. Solve $x + \frac{1}{3} = \frac{5}{6}$

ANSWER $\frac{1}{2}$

6. Solve $x - \frac{2}{5} = \frac{1}{10}$

ANSWER $\frac{1}{2}$

7. $\frac{3}{8} + \frac{1}{8} - \frac{1}{4}$

ANSWER $\frac{1}{4}$

Medium · the standard question

7 PROBLEMS

1. $\frac{1}{2} - \frac{2}{3} + \frac{3}{4}$

ANSWER $\frac{7}{12}$

2. $\frac{2}{5} + \frac{1}{2} - \frac{7}{10}$

ANSWER $\frac{1}{5}$

3. $-\frac{1}{3} - \frac{1}{4} + \frac{5}{6}$

ANSWER $\frac{1}{4}$

4. $0.25 - \frac{1}{3}$

ANSWER $-\frac{1}{12}$

5. $\frac{1}{3} + 0.2$

ANSWER $\frac{8}{15}$

6. Solve $x + \frac{3}{4} = \frac{1}{4}$

ANSWER $-\frac{1}{2}$

7. $1 - \frac{1}{2} - \frac{1}{3}$

ANSWER $\frac{1}{6}$

Hard · stretch and reason

7 PROBLEMS

1. $\frac{1}{2} - \frac{1}{3} + \frac{1}{4} - \frac{1}{6}$

ANSWER $\frac{1}{4}$

2. $-\frac{3}{5} + 0.25 - \frac{1}{4}$

ANSWER $-\frac{3}{5}$

3. $2\frac{1}{3} - 1\frac{1}{2} + \frac{5}{6}$

ANSWER $1\frac{2}{3}$

4. Solve $\frac{2}{3} - x = -\frac{1}{6}$

ANSWER $\frac{5}{6}$

5. Three pieces of a rod are $\frac{1}{4}$, $\frac{1}{3}$, $\frac{1}{6}$ of it: the rest

ANSWER $\frac{1}{4}$

6. $0.75 - \frac{2}{3} + \frac{1}{12}$

ANSWER $\frac{1}{6}$

7. The mean of $\frac{1}{2}$, $\frac{1}{3}$, $\frac{1}{6}$

ANSWER $\frac{1}{3}$

07 Homework

Homework — 5 tasks

Use one common denominator for each whole chain, and check every equation.

1. Compute $\frac{2}{3} - \frac{1}{2} + \frac{1}{6}$.
2. Compute $-\frac{1}{4} + \frac{2}{5} - \frac{1}{10}$.
3. Compute $0.4 + \frac{1}{3}$.
4. Solve $x + \frac{5}{6} = \frac{1}{3}$.
5. Solve $\frac{3}{4} - x = \frac{1}{8}$.